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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

RI2102-1 – CONTROL ENGINEERING

(For RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: i) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

ii) **Require 2 Normal graph sheets.**

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- An open-loop control system is characterized by:**
 - No feedback
 - Feedback presence
 - Nonlinear control
 - Adaptive behavior
- In a closed-loop control system, the output is compared with:**
 - Disturbance
 - Setpoint or Reference input
 - Amplifier gain
 - Actuator position
- The equation $F = ma$ is used in:**
 - Electrical systems
 - Translational mechanical systems
 - Rotational mechanical systems
 - Hydraulic systems
- In block diagram algebra, blocks in series are combined by:**
 - Addition
 - Multiplication
 - Subtraction
 - Division
- Feedback-loop gain is calculated as:**
 - $G - H$
 - $G \times H$
 - $G / (1 + GH)$
 - $1 + GH$
- Two blocks with gains $G_1 = 4$ and $G_2 = 2$ are in parallel. What is the equivalent**
 - 8
 - 6
 - 2
 - 4
- Find the equivalent gain of a system with $G_1 = 5$, $G_2 = 3$ in series.**
 - 15
 - 2
 - 8
 - 1
- A block diagram has $G = 5$ and $H = 0.2$. What is the closed-loop transfer function?**
 - $5/(1 - 1)$
 - $5/(1 + 1)$
 - $5/0.8$
 - $1/5$
- Which of the following is NOT a standard test signal?**
 - Unit Step
 - Impulse
 - Parabolic
 - Sinusoidal
- What is the Laplace transform of a unit step function?**
 - $1/s$
 - s
 - 1
 - s^2

11. **Routh-Hurwitz criterion is used to determine:**
 - A) Phase margin
 - B) Poles of the system
 - C) Gain margin
 - D) Stability of the system
12. **The gain crossover frequency is where:**
 - A) Gain = 0 dB
 - B) Phase = 180 degrees
 - C) System becomes unstable
 - D) Poles cross into right half-plane
13. **A system is stable if all its poles are in:**
 - A) Right half-plane
 - B) Left half-plane
 - C) Imaginary axis
 - D) On the x axis
14. **Which type of system has zero steady-state error for a step input?**
 - A) Type 0
 - B) Type 1
 - C) Type 2
 - D) Type 3
15. **The Nyquist plot encircles (-1,0) in clockwise direction. What does this indicate?**
 - A) System is stable
 - B) System is unstable
 - C) System has zero gain
 - D) System is marginally stable
16. **A system with positive phase margin is:**
 - A) Stable
 - B) Marginally stable
 - C) Unstable
 - D) Critically stable
17. **How many branches does the root locus have?**
 - A) Infinite
 - B) Equal to the number of open-loop zeros
 - C) Equal to the sum of open-loop poles and zeros
 - D) Equal to the number of open-loop poles
18. **Asymptotes in a root locus indicate:**
 - A) The directions in which the root locus branches tend to infinity
 - B) The open-loop gain of the system
 - C) The frequency response of the system
 - D) The phase margin of the system
19. **What happens when the root locus crosses the imaginary axis?**
 - A) The system becomes unstable
 - B) The system remains stable
 - C) The system enters into oscillations
 - D) The gain reduces to zero
20. **The output equation in state-space representation is:**
 - A) $y = A^2 x + Bu$
 - B) $y = Ax + Bu$
 - C) $y = dx/dt$
 - D) $y = Cx + Du$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

1. a) With examples explain closed loop and open loop systems.
b) Find the transfer function $\theta_1(s)/T(s)$ and $\theta_2(s)/\theta_1(s)$ for the translational mechanical system shown in figure.1(b).

Marks	BT*	CO*	PO*
8	L*2	1	1,2

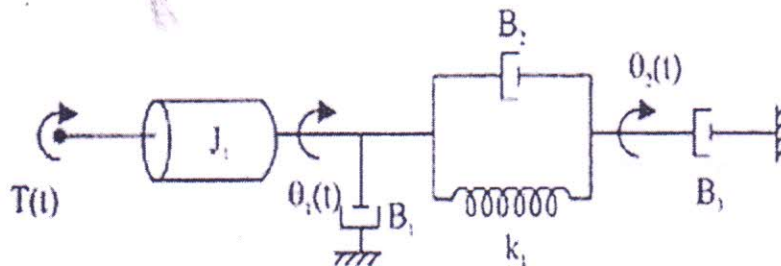


Figure 1(b)

8	L3	2	1,2
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2. a) Derive the transfer function of armature controlled DC motor.

6	L2	1	1,2
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- b) Find the transfer function by reducing the block diagram shown in figure. 2(b).

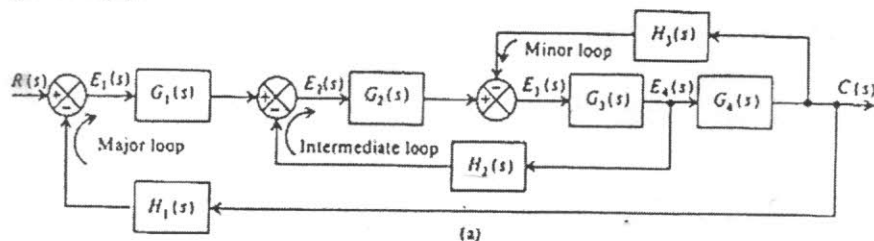


Figure. 2(b)

10 L3 2 1,2

6 L2 1 1,2

3. a) Write a short note on lag, lead lag compensators.
b) Define Mason's gain formula and find the transfer function by block diagram reduction for the signal flow graph shown in fig 3(b)

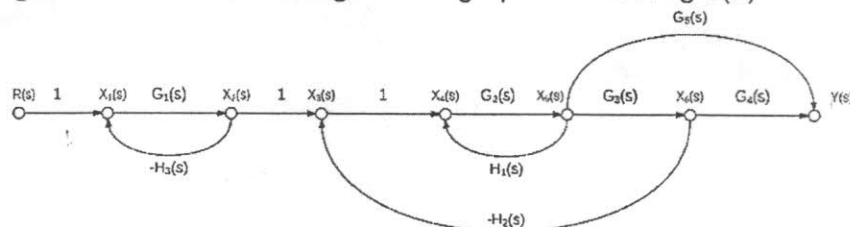


Figure. 3(b)

10 L3 2 1,2

Unit – II

4. a) Define stability and hence stable, unstable, marginally stable and conditional stability of a unity feedback system.
b) For a certain control system $G(s)H(s) = K/[s(s+2)(s+10)]$ sketch the Nyquist plot and hence calculate the range of values of K for stability.

6 L2 3 1,2

10 L3 4 1,2

5. a) Starting from the output system $C(t)$ derive expressions for:

- i) Rise time (T_r)
ii) Peak time (T_p).

Of an under damped second order system subjected to unit step input.

8 L2 3 1,2

- b) Check the stability of the given characteristic equation using Routh's method $s^6 + 3s^5 + 5s^4 + 9s^3 + 8s^2 + 6s + 4 = 0$.

8 L3 4 1,2

6. a) Derive the expression for a time response of a second order system subjected to a unit step input for $\zeta < 1$.

8 L2 3 1,2

- b) A unity feedback system has $G(s) = 20(1+s)/s^2(2+s)(4+s)$, calculate its steady state error coefficients and error when the applied input $r(t) = 40 + 2t + 5t^2$.

8 L3 4 1,2

Unit – III

7. a) Write advantages and disadvantages transfer function.
b) Sketch the complete root locus of the system having $G(s)H(s) = K/[s(s+1)(s+2)(s+3)]$.

4 L2 5 1,2

12 L3 5 1,2

8. a) Define and explain i) state variables ii) state vector
b) $G(s)H(s) = K/[s(s+3)(s+5)]$. Draw root locus determine Gain K for damping ratio = 0.6.

4 L2 5 1,2

12 L3 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
